ST2334 Cheat Sheet, AY26/27 S1

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1 Basic Concepts of Probability

1.1 Experiments, sample spaces, and events

- **Statistical Experiment** is a procedure that produces an observation.
- **Sample Space** is the set of all possible outcomes of a statistical experiment.
- **Event** is a subset of the sample space.
- **Sample Point** is a single outcome of a statistical experiment.

$$\text{Sample Point} \subseteq \text{Event} \subseteq \text{Sample Space}$$

Sure Event and Null Event

- **Sure Event** = Sample Space, S
- **Null Event** = Empty Set / Complement of Sure Event

1.2 Event Operations & Relationships

- **Union** of A and B, $(A \cup B)$

$$A \cup B = \{x \in U \mid x \in A \vee x \in B\}$$

- **Intersection** of A and B, $(A \cap B)$

$$A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$$

- **Difference** of B minus A, $(B - A)$

$$B - A = \{x \in U \mid x \in B \wedge x \notin A\} = B \cap \bar{A}$$

- **Complement** of A $(\bar{A})$

$$\bar{A} = \{x \in S \mid x \notin A\}$$

- **Mutually Exclusive or Disjoint** Events

$$A \cap B = \emptyset$$

Contained and Equivalent

- **Contained** $\subset$

$$A \subset B \Leftrightarrow$$
$$\forall x, x \in A \Rightarrow x \in B$$

- **Equivalent** $=$

$$A = B \Leftrightarrow$$
$$A \subset B \wedge B \subset A$$

Subset Relations

No.	Law	Identities
1	Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
2	Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
3	Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4	Identity laws	$A \cap U = A$ $A \cap \emptyset = \emptyset$ $A \cup U = U$ $A \cup \emptyset = A$
5	Complement laws	$\overline{\overline{A}} = A$ $A \cap \overline{A} = \emptyset$ $\overline{U} = \emptyset$
6	Idempotent laws	$A \cup A = A$ $A \cap A = A$
7	De Morgan's laws	$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$
8	Absorption laws	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

1.3 Counting Methods

Multiplication and Addition Principles

- **Multiplication Principle**

A sequence of r stages with n_1, n_2, …, n_r choices has n_1n_2…n_r outcomes.

$$n_1 \times n_2 \times \cdot \times n_r = n_1 n_2 ... n_r$$

- **Addition Principle**

k non-overlapping procedures with $n_1, n_2, ..., n_k$ outcomes have $n_1 + n_2 + ... + n_k$ outcomes in total.

$$n_1 + n_2 + ... + n_k = n_1 + n_2 + ... + n_k$$

Factorial, Permutations, and Combinations

- **Factorial** $(n!)$

$$n! = n(n-1)(n-2)...2 \cdot 1$$

- **Permutation** (P_r^n)

Selection and arrangement of r objects from n objects, where order matters.

$$P_r^n = \texttt{frac}\{n!\}\{(n-r)!\}$$

- **Combination** $(\binom{n}{r})$

Selection of r objects from n objects, where order does not matter.

$$\binom{n}{r} = \texttt{frac}\{n!\}\{r!(n-r)!\}$$

1.4 Probability

Probability of an event A is a measure of the likelihood of the event occurring.

Relative Frequency

Suppose an experiment is repeated n times and event A occurs m times. The relative frequency of A is given by:

$$P(A) = \lim_{\{n \to \infty\}} \texttt{frac}\{m\}\{n\}$$

As n increases, the relative frequency of A approaches the probability of A.

Axiomatic Definition of Probability

A probability function P is a function that assigns a number P(A) to each event A in the sample space S, such that:

$$0 \leq P(A) \leq 1$$

$$P(S) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Properties of Probability

Prop.	Name	Statement
1.4	Empty set	$P(\emptyset) = 0$
1.5	Finite additivity	If $A_i \cap A_j = \emptyset$ for $i \neq j$, then $P(A_1 \cup ... \cup A_n) = P(A_1) + ... + P(A_n)$
1.6	Complement	$P(\bar{A}) = 1 - P(A)$
1.7	Partition of A	$P(A) = P(A \cap B) + P(A \cap \bar{B})$
1.8	Inclusion-exclusion	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
1.9	Monotonicity	$A \subset B \Rightarrow P(A) \leq P(B)$
1.10	Union bound (Boole's inequality)	$P(A_1 \cup ... \cup A_n) \leq P(A_1) + ... + P(A_n)$